

When Do Factor Premia Break?

A Pre-Registered Early-Warning Study of Structural Instability in Equity Factor Returns

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Abstract

This paper asks whether changes in the recent return distribution of an equity factor provide advance warning of a subsequent collapse in that factor’s premium. The study is designed as an event-detection problem rather than a trading-strategy backtest. Before examining holdout results, the factor universe, collapse definition, development case, candidate diagnostics, evaluation horizon, and multiple-testing plan were pre-registered. Daily U.S. factor returns for the market, size, value, profitability, investment, and momentum factors are examined from July 1963 through May 2026. Collapse episodes are defined mechanically as sustained breaches of a trailing Sharpe-ratio boundary. Three causal diagnostics—rolling skewness, rolling excess kurtosis, and autocorrelation of squared returns—are developed only on a single Momentum episode, after which the selected specification is frozen.

The development exercise selects a 63-day excess-kurtosis statistic standardized against its prior 252 observations, with a warning threshold of two historical standard deviations. On 86 holdout collapse episodes, the frozen detector identifies 42 events, for recall of 48.8%, warning precision of 7.8%, and a median lead time of 66 trading days. Factor-stratified bootstrap intervals are 38.4–59.3% for recall, 5.6–10.1% for precision, and 46–91 days for median lead time. A circular-shift null preserves each factor’s warning frequency and temporal clustering; observed recall and precision exceed their null means only modestly, with one-sided permutation p -values of 0.164. On the common VIX-available sample, the kurtosis detector is more selective than a causal VIX-level baseline but has lower recall. The two signals detect different subsets of events in the common sample. The evidence therefore supports excess kurtosis as a potentially informative but noisy fragility indicator, not as a validated standalone collapse classifier.

Keywords: equity factors; factor premia; momentum crashes; excess kurtosis; early-warning systems; pre-registration; data leakage; out-of-sample validation

JEL classifications: C12, C52, G11, G12

1 Introduction

Equity factor premia are often presented as persistent compensation for systematic risk or as durable consequences of behavioral and institutional frictions. In practice, however, factor strategies can experience long episodes in which their historical premium weakens sharply, reverses, or becomes unusually unstable. Momentum is the clearest example: despite a strong long-run average return, it can suffer abrupt losses during market rebounds following severe declines (Daniel and Moskowitz, 2016). Value, size, profitability, and investment factors have also experienced extended periods of disappointing or negative realized performance.

The practical question is not merely whether factor premia vary over time. It is whether an investor could have observed a factor’s own recent return distribution and obtained a useful warning before a formally defined collapse. This distinction matters. An explanation produced after a drawdown can be economically persuasive while offering no information that would have been available in real time. Similarly, an indicator can appear successful if its parameters, event dates, or evaluation horizon are selected after viewing the full sample. These risks are especially important in empirical asset pricing, where repeated specification search and multiple testing can generate fragile findings (Harvey et al., 2016).

This paper imposes a deliberately strict research design. The study follows a pre-specified and repository-committed design, uses only trailing statistics, defines collapse events mechanically, permits parameter development on one Momentum episode only, freezes the resulting detector before holdout evaluation, reports all attempted diagnostics, and tests future-data invariance programmatically. The detector is evaluated using precision, recall, false-positive frequency, and lead time rather than portfolio returns. This design shifts the focus from whether a signal can be made profitable in sample to whether it can reliably distinguish fragile from non-fragile periods.

The empirical analysis uses daily U.S. factor returns for the market excess return, SMB, HML, RMW, CMA, and Momentum. These series extend the market, size, and value framework of Fama and French (1993), the momentum factor widely associated with Carhart (1997), and the profitability and investment factors of Fama and French (2015). A collapse is defined as a five-day-confirmed breach in which the trailing 252-day annualized Sharpe ratio falls more than two standard deviations below its prior 1,260-day Sharpe distribution. Implementing both the pre-specified five-day entry and five-day recovery-confirmation rules produces 87 episodes, of which one Momentum episode beginning October 7, 2009 is reserved for development and 86 are retained for holdout testing.

Three diagnostics are considered on the development case: 63-day rolling skewness, 63-day rolling excess kurtosis, and 63-day lag-one autocorrelation of squared returns. Each diagnostic is standardized using only its prior 252 valid values. Only excess kurtosis produces a warning before the development collapse at more than one adjacent candidate threshold. The final frozen rule warns when the excess-kurtosis historical z -score exceeds 2.0. It produces a development warning on September 17, 2009, 14 trading days before the mechanically dated collapse.

The holdout results are mixed. The frozen detector identifies 42 of 86 collapses, with recall of 48.8%, but issues 536 distinct warning events, implying precision of 7.8% and an unmatched-warning

rate of 92.2%. When successful, warnings are meaningfully early: the median lead time is 66 trading days. A factor-stratified bootstrap indicates substantial uncertainty, while a circular-shift null suggests that the observed alignment is not statistically unusual at conventional levels. No individual factor remains significant after Benjamini–Hochberg or Bonferroni correction.

A common-sample comparison with VIX provides additional context. From December 1990 through May 2026, the frozen kurtosis detector identifies 31 of 55 collapses with 298 factor-specific warning events, while a causal VIX-level z -score identifies 35 events with 540 factor-specific warning events. Kurtosis therefore has lower recall but higher precision. Importantly, 11 collapses are detected only by kurtosis and 15 only by VIX, showing that the two signals detect different subsets of events and suggesting—but not statistically establishing—that factor-specific tail instability may contain information distinct from broad market volatility.

The paper’s contribution is methodological and empirical. Methodologically, it provides a transparent example of pre-registration, causal computation, parameter freezing, leakage testing, and honest null reporting in a small-sample financial detection problem. Empirically, it shows that excess kurtosis is associated with a nontrivial fraction of later factor collapses and produces useful lead times when correct, but is too noisy to function as a reliable standalone warning system. The evidence is better interpreted as a weak signal worthy of further study than as a deployable forecasting model.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data and research design. Section 4 defines collapse episodes. Section 5 describes diagnostic development and parameter freezing. Section 6 presents holdout results. Section 7 reports statistical validation. Section 8 compares the detector with VIX and presents sensitivity checks. Section 10 discusses limitations, and Section 11 concludes.

2 Related Literature

2.1 Equity factor models

The factor universe is rooted in the standard empirical asset-pricing literature. Fama and French (1993) show that market, size, and book-to-market factors capture substantial common variation in stock returns. Carhart (1997) adds a momentum factor to the three-factor model in a performance-attribution setting. Fama and French (2015) later add profitability and investment factors, producing the market, SMB, HML, RMW, and CMA structure used in this study. The present paper does not propose a new priced factor. Instead, it studies the time-series stability of these established factor returns.

2.2 Factor crashes and conditional risk

Momentum crashes motivate the development design. Daniel and Moskowitz (2016) document infrequent but severe momentum losses that occur in panic states, after market declines, during periods of high volatility, and alongside sharp market rebounds. Their results imply that the

conditional distribution of momentum returns can change meaningfully across regimes. Related work on volatility-managed portfolios shows that scaling exposure by estimated variance can materially change factor performance (Barroso and Santa-Clara, 2015; Moreira and Muir, 2017). These studies focus primarily on conditional portfolio construction and risk management. The present paper instead asks whether simple distributional diagnostics can classify an approaching collapse before it begins under a separately fixed event definition.

2.3 Research design and multiple testing

Empirical finance faces a severe multiple-testing problem. Harvey et al. (2016) argue that conventional significance thresholds are inadequate after extensive factor search. Backtest overfitting can similarly create impressive in-sample performance that fails out of sample (Bailey et al., 2017). These concerns motivate the design used here: one development event, an explicit holdout, disclosure of failed diagnostics, frozen parameters, causal-only computation, and multiple-testing adjustments. The paper does not claim that these protections eliminate all researcher degrees of freedom. They do, however, make those choices visible and constrain adaptation after holdout results are observed.

3 Data and Pre-Registered Design

3.1 Data

Daily factor returns are obtained from the Kenneth R. French Data Library (French, 2026). The sample contains 15,833 observations from July 1, 1963 through May 29, 2026. The five-factor file supplies the market excess return (MKT–RF), SMB, HML, RMW, CMA, and the risk-free rate. A separate daily Momentum file supplies MOM. Published percentage returns are converted to decimal form. The cleaned data are stored in Parquet format; raw source archives and SHA-256 checksums are retained for reproducibility.

Daily VIX closing levels are obtained from Cboe (Cboe Global Markets, 2026) and used only as an external baseline, not as an input to the factor-specific detector. Because VIX begins later than the factor sample and requires a 252-observation historical standardization window, baseline comparisons use a common evaluation sample beginning December 31, 1990 and ending May 29, 2026.

3.2 Pre-registration and development split

The research plan was written and committed to the project repository before diagnostic development. It fixed the six factors, three candidate diagnostics, objective collapse rule, Momentum development case, holdout structure, 126-trading-day warning horizon, five-day warning reset, causal-only requirement, and multiple-testing plan. The intended historical development context was the 2008–2009 Momentum crash. The objective event rule, however, did not date the nearby Momentum



Figure 1: Research design and evaluation sequence. The detector is developed on one event and frozen before holdout evaluation.

deterioration until October 7, 2009. That mechanically generated date is retained rather than adjusted to better match the narrative crash date.

All factors other than the single October 7, 2009 Momentum event are held out. Later Momentum events are also included in the holdout set. No factor-specific retuning is permitted after the frozen detector is saved.

3.3 Causal-only computation

For every computed statistic, the value dated t is a function only of observations dated t or earlier. Historical means and standard deviations are estimated from values through $t - 1$. No centered windows or full-sample normalization are used. Synthetic leakage tests modify all observations after a chosen cutoff and confirm that outputs through the cutoff remain unchanged. Historical collapse labels use future confirmation only to determine whether a below-threshold or recovery run persists for five days; those labels are treated as ex post ground truth, not as live signals.

4 Objective Collapse Definition

For factor i , let $r_{i,t}$ denote its daily return. The trailing 252-day annualized Sharpe ratio is

$$SR_{i,t} = \sqrt{252} \frac{\bar{r}_{i,t}^{(252)}}{s_{i,t}^{(252)}}, \quad (1)$$

where the mean and sample standard deviation use dates $t - 251$ through t .

The causal five-year reference mean and standard deviation are

$$\mu_{i,t}^{SR} = \text{mean}(SR_{i,t-1260}, \dots, SR_{i,t-1}), \quad (2)$$

$$\sigma_{i,t}^{SR} = \text{sd}(SR_{i,t-1260}, \dots, SR_{i,t-1}). \quad (3)$$

The collapse boundary is

$$B_{i,t} = \mu_{i,t}^{SR} - 2\sigma_{i,t}^{SR}. \quad (4)$$

A collapse begins on the first day of a sequence in which $SR_{i,t} < B_{i,t}$ for at least five consecutive trading days. The start date is recorded retroactively as the first day of the qualifying sequence. A

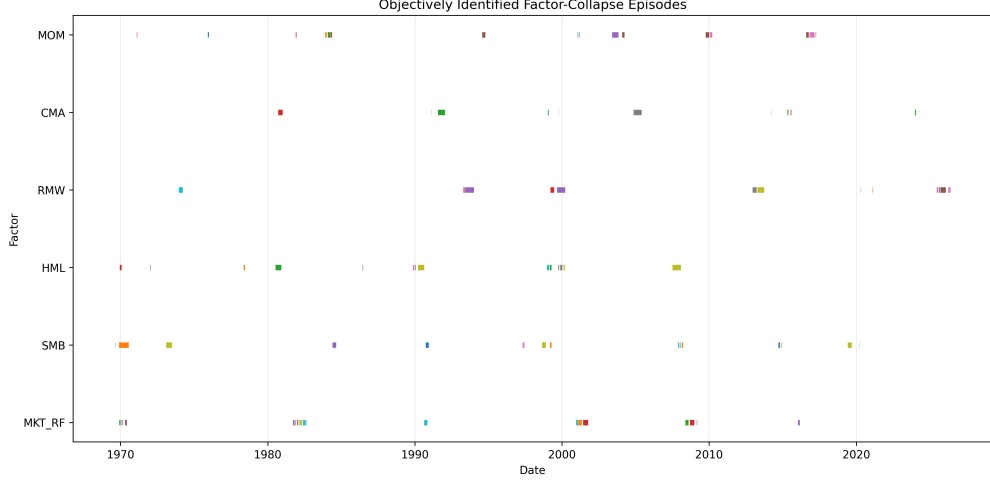


Figure 2: Mechanically dated factor-collapse episodes. Horizontal segments indicate each episode’s start and end dates.

collapse ends immediately before a recovery sequence in which the Sharpe ratio remains above the boundary for at least five consecutive trading days.

The rule identifies 87 total episodes. One Momentum episode is reserved for development, leaving 86 holdout episodes: 17 for MKT–RF, 16 for SMB, 14 for HML, 13 for RMW, 11 for CMA, and 15 for MOM.

The rule should be interpreted as a measure of sustained premium deterioration, not necessarily as a timestamp for the first extreme daily loss. Its failure to date the main 2008 Momentum crash at its onset is therefore an important limitation rather than a reason for post-hoc adjustment.

5 Diagnostic Development and Parameter Freeze

5.1 Candidate diagnostics

Three 63-trading-day diagnostics are evaluated on the development sample:

- (i) sample skewness of daily factor returns;
- (ii) Fisher excess kurtosis of daily factor returns; and
- (iii) lag-one autocorrelation of squared daily returns.

For each raw diagnostic D_t , the causal standardized value is

$$Z_t = \frac{D_t - \bar{D}_{t-1}^{(252)}}{s_{D,t-1}^{(252)}}, \quad (5)$$

where the reference window contains the prior 252 valid diagnostic observations. Warning direction is standardized so that larger values imply greater fragility: skewness is multiplied by -1 , while

excess kurtosis and squared-return autocorrelation retain their signs.

Candidate thresholds are 1.5, 2.0, and 2.5. A new warning event is recorded on the first threshold-crossing date after at least five consecutive non-warning days. A warning is associated with a collapse if the collapse begins within the next 126 trading days. When multiple unmatched warnings fall within an event’s attribution window, the earliest warning is assigned to the event. This convention emphasizes the maximum available advance notice under the frozen matching rule. The combination rule “at least two of three diagnostics exceed the same threshold” is also evaluated.

5.2 Development result

Skewness and squared-return autocorrelation fail to warn before the October 7, 2009 development event at every candidate threshold. The two-of-three combination rule also fails at all thresholds. Excess kurtosis detects the event at thresholds 1.5 and 2.0, but not at 2.5. At threshold 1.5, the first qualifying warning is July 30, 2009, 48 trading days before the collapse. At threshold 2.0, the first warning is September 17, 2009, 14 trading days before the collapse.

The threshold of 2.0 is selected because it remains interpretable as an unusually large historical deviation, detects the event, and issues fewer earlier warning events than 1.5. The frozen specification is therefore a 63-day excess-kurtosis statistic, standardized using the prior 252 observations, with a warning threshold of 2.0 and a five-day reset rule.

Table 1: Development-case diagnostic results

Diagnostic	Threshold	Detected	Lead time	Total warnings
Skewness	1.5	No	–	75
Skewness	2.0	No	–	50
Skewness	2.5	No	–	30
Excess kurtosis	1.5	Yes	48	80
Excess kurtosis	2.0	Yes	14	71
Excess kurtosis	2.5	No	–	62
Squared-return ACF	1.5	No	–	92
Squared-return ACF	2.0	No	–	67
Squared-return ACF	2.5	No	–	50

Notes: Lead time is measured in trading days. Warning counts cover the development sample. The selected frozen specification is excess kurtosis at threshold 2.0. The two-of-three combination rule failed at all three thresholds and is omitted from the body of the table.

6 Holdout Results

6.1 Pooled performance

The frozen detector is applied without modification to 86 holdout collapse episodes. It detects 42 events and misses 44. Across the full holdout history, it produces 536 distinct factor-specific warning

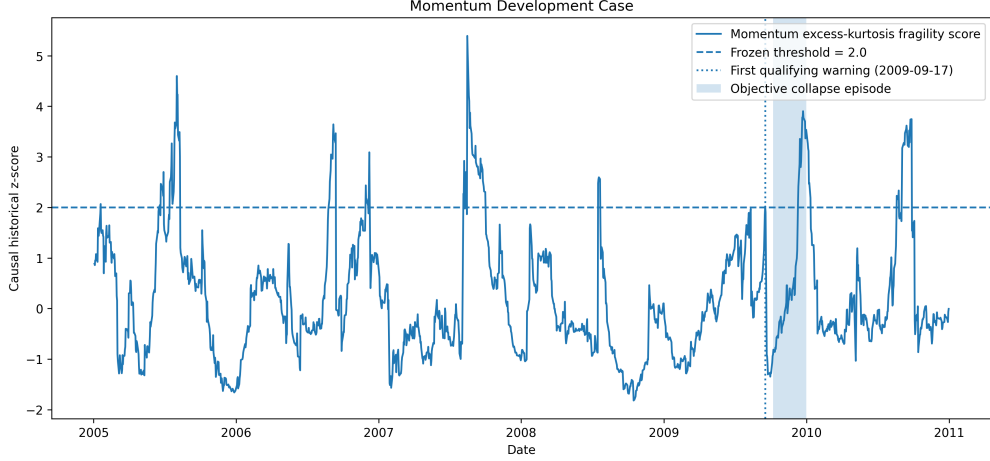


Figure 3: Momentum development case. The plot shows the causal excess-kurtosis fragility score, frozen threshold, first qualifying warning, and objective collapse period.

events, of which 42 are matched one-to-one to collapses. The resulting recall is 48.8%, precision is 7.8%, and the unmatched-warning rate is 92.2%. Among detected events, median lead time is 66 trading days and mean lead time is 66.8 days.

Table 2: Primary holdout results

Metric	Estimate
Holdout collapse events	86
Detected events	42
Missed events	44
Warning events	536
True-positive warnings	42
False-positive warnings	494
Recall	0.488
Precision	0.078
Unmatched-warning rate	0.922
Median lead time	66 trading days
Mean lead time	66.8 trading days

Notes: Precision is measured at the warning-event level; recall is measured at the collapse-event level. Matching is one-to-one within factor and uses the frozen 126-trading-day horizon. The unmatched-warning rate is the proportion of warning events not assigned to a collapse within that horizon; it is not the conventional classification false-positive rate, which would require a defined count of true-negative opportunities.

The detector therefore exhibits moderate event coverage but low selectivity. It is better characterized as a high-frequency fragility alarm than as a precise collapse classifier. The low precision is not primarily an artifact of a narrowly missed attribution horizon. An exploratory

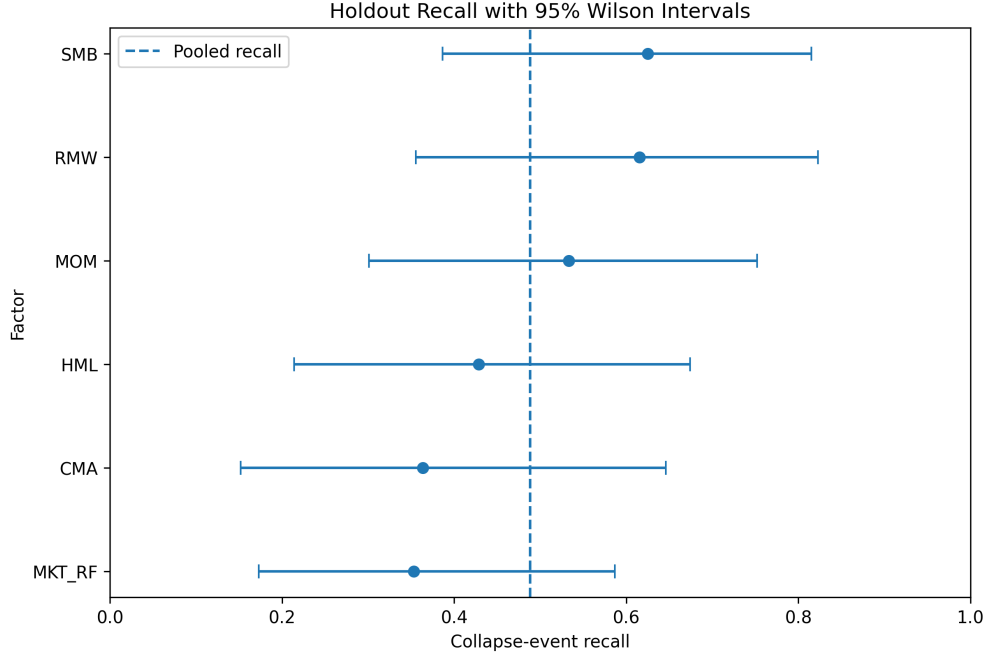


Figure 4: Holdout recall by factor with 95% Wilson intervals. The dashed line marks pooled recall.

appendix classification finds that only 23 additional warnings occur 127–252 trading days before a later collapse, while 377 occur more than one trading year before the next collapse and are too early to attribute meaningfully.

6.2 Factor-level heterogeneity

Table 3 reports factor-level performance. Recall is highest for SMB (62.5%) and RMW (61.5%), followed by Momentum (53.3%). It is lowest for MKT–RF (35.3%) and CMA (36.4%). Precision remains low for every factor, ranging from 4.7% for CMA to 11.4% for SMB.

Table 3: Holdout performance by factor

Factor	Events	Detected	Recall	Precision	Median lead	Warnings/year
MKT–RF	17	6	0.353	0.076	64.0	1.26
SMB	16	10	0.625	0.114	78.0	1.40
HML	14	6	0.429	0.067	72.5	1.43
RMW	13	8	0.615	0.080	51.0	1.59
CMA	11	4	0.364	0.047	78.5	1.35
MOM	15	8	0.533	0.085	65.5	1.49

Notes: Median lead is measured in trading days for detected events. Warnings per year use the factor sample length.

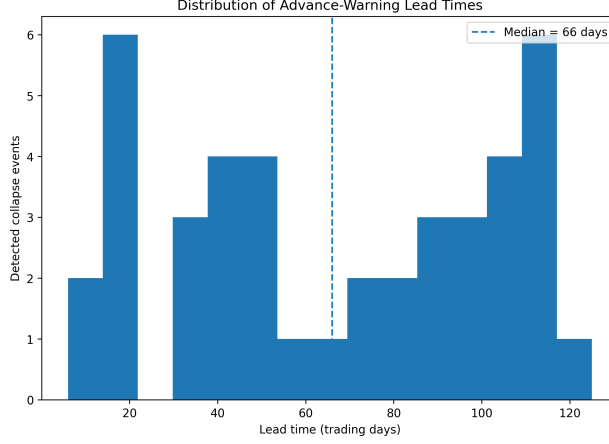


Figure 5: Distribution of lead times among detected holdout collapse events.

7 Statistical Validation

7.1 Bootstrap uncertainty

A factor-stratified nonparametric bootstrap resamples event outcomes, warning classifications, and detected-event lead times within each factor. With 10,000 replications, the 95% percentile interval for recall is 38.4–59.3%, the interval for precision is 5.6–10.1%, and the interval for median lead time is 46–91 trading days.

Table 4: Bootstrap confidence intervals

Metric	Estimate	2.5th percentile	97.5th percentile
Recall	0.488	0.384	0.593
Precision	0.078	0.056	0.101
Median lead time (days)	66	46	91

The bootstrap is stratified by factor but does not fully model serial dependence between nearby warnings or collapses. The intervals should therefore be interpreted as descriptive sampling uncertainty under within-factor resampling, not as dependence-robust asymptotic inference.

7.2 Circular-shift null

The null benchmark circularly shifts each factor’s complete warning-date pattern through the trading calendar. This preserves the number of warnings, their spacing, and their clustering while breaking the observed alignment with collapse dates. Five thousand simulations are evaluated using the same one-to-one matching rule.

Observed recall is 0.488, compared with a null mean of 0.429 and null 95th percentile of 0.523. The one-sided permutation p -value is 0.164. Observed precision is 0.078, compared with a null mean of 0.069 and null 95th percentile of 0.084; its permutation p -value is also 0.164.

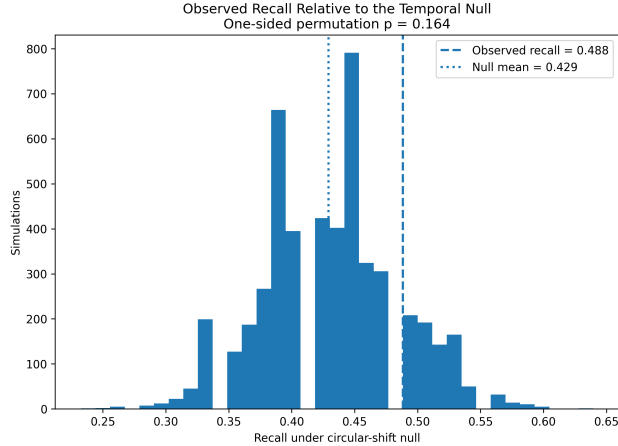


Figure 6: Observed holdout recall relative to the circular-shift null distribution.

Table 5: Circular-shift null comparison

Metric	Observed	Null mean	Null median	Null 95th pct.	p -value
Recall	0.488	0.429	0.430	0.523	0.164
Precision	0.078	0.069	0.069	0.084	0.164

Because each simulation preserves both the total number of collapse events and the total number of warning events, recall and precision are monotonic transformations of the same matched-event count. Their permutation rankings, and therefore their one-sided permutation p -values, are identical by construction.

The detector thus performs modestly better than the average temporally displaced warning pattern, but the difference is not statistically significant at conventional levels. The results do not justify a claim that the detector “beats chance” under this null.

7.3 Factor-level inference and multiple testing

No individual factor is statistically significant after correction. SMB has the strongest raw result, with recall of 0.625 versus a null mean of 0.423 and a raw permutation p -value of 0.087. Its Benjamini–Hochberg and Bonferroni adjusted p -values are both 0.522. All other adjusted p -values exceed 0.65 under Benjamini–Hochberg and equal 1.0 under Bonferroni except SMB.

Table 6: Factor-level statistical validation

Factor	Recall	95% CI low	95% CI high	Null mean	Raw p	BH adjusted	Bonferroni
MKT-RF	0.353	0.173	0.587	0.348	0.573	0.688	1.000
SMB	0.625	0.386	0.815	0.423	0.087	0.522	0.522
HML	0.429	0.214	0.674	0.421	0.565	0.688	1.000
RMW	0.615	0.355	0.823	0.472	0.217	0.652	1.000
CMA	0.364	0.152	0.646	0.452	0.818	0.818	1.000
MOM	0.533	0.301	0.752	0.483	0.441	0.688	1.000

These results are consistent with either a weak signal or limited power. The number of collapse episodes per factor ranges from 11 to 17, so moderately large true differences can remain statistically unresolved.

8 VIX Baseline and Robustness

8.1 Common-sample VIX comparison

The VIX baseline is the causal historical z -score of the daily VIX closing level relative to its prior 252 observations. A warning begins when the score exceeds 2.0 and follows the same five-day reset and 126-day event-matching rules. Both VIX and kurtosis are evaluated from December 31, 1990 through May 29, 2026, covering 55 holdout collapse episodes.

The frozen kurtosis detector identifies 31 events, for recall of 56.4%, while issuing 298 factor-specific warnings and achieving precision of 10.4%. VIX identifies 35 events, for recall of 63.6%, while issuing 540 factor-specific warnings and achieving precision of 6.5%. VIX therefore provides higher event coverage and longer median lead time, but at a substantially larger warning burden and lower precision.

Table 7: Common-sample comparison with VIX

Detector	Events	Detected	Warnings	Recall	Precision	Median lead
Frozen excess kurtosis	55	31	298	0.564	0.104	83
VIX level z -score	55	35	540	0.636	0.065	103

Notes: The sample runs from December 31, 1990 through May 29, 2026. Warning counts are factor-specific; the same market-wide VIX warning date can appear once for each factor. Across unique VIX dates, 28 of 90 warning dates are matched to at least one factor collapse.

8.2 Detection overlap

The two detectors identify different subsets of events. Of 55 common-sample collapses, 20 are detected by both, 11 only by kurtosis, 15 only by VIX, and 9 by neither. Kurtosis is therefore not merely reproducing a generic market-volatility signal. Conversely, VIX detects events not captured

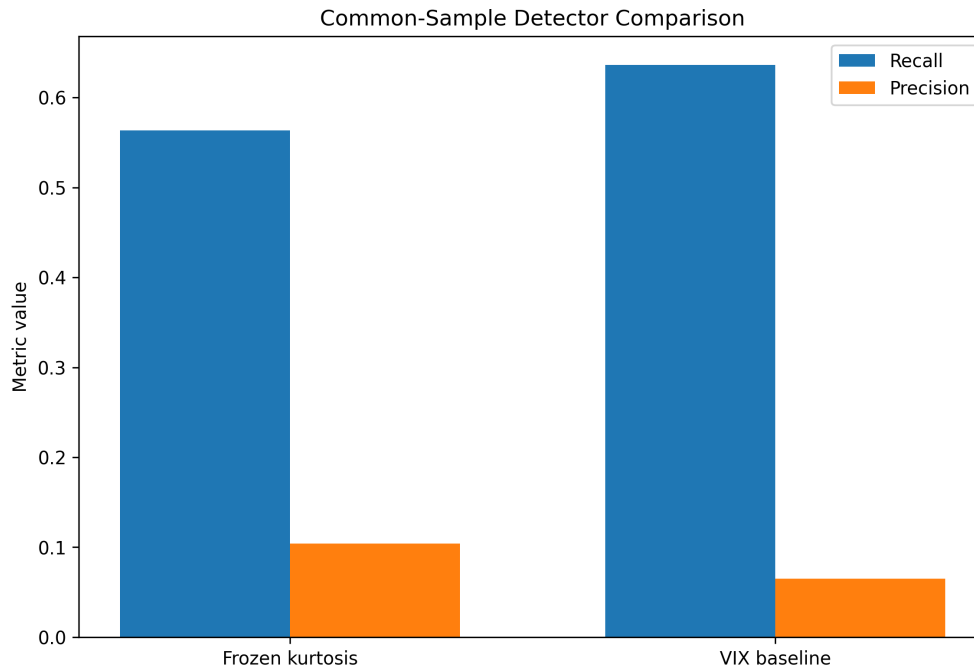


Figure 7: Common-sample recall and precision for the frozen excess-kurtosis detector and causal VIX baseline.

by factor-specific excess kurtosis. This pattern suggests—but does not statistically establish—that the signals may contain distinct information. A combined detector was not pre-specified or formally evaluated.

Table 8: Overlap in event detection

Classification	Event count
Detected by both	20
Kurtosis only	11
VIX only	15
Detected by neither	9

8.3 Sensitivity to warning horizon

Holding warning dates fixed, extending the event-attribution horizon from 63 to 252 trading days raises measured recall from 32.6% to 67.4% and precision from 5.2% to 10.8%. Median lead time rises from 32.5 to 116 days. The pre-registered 126-day horizon lies between these cases. The result demonstrates that headline classification metrics are materially affected by the attribution convention, even though the detector itself is unchanged.

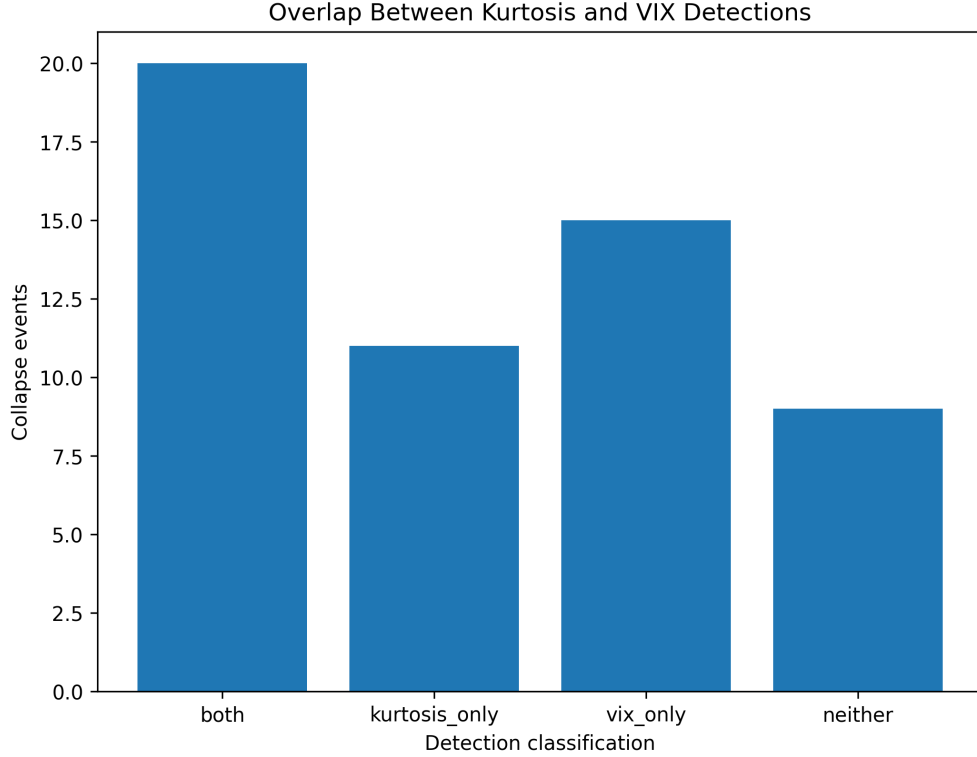


Figure 8: Overlap between events detected by excess kurtosis and by the VIX baseline.

Table 9: Sensitivity to the event-attribution horizon

Horizon (days)	Recall	Precision	Median lead
63	0.326	0.052	32.5
126	0.488	0.078	66.0
252	0.674	0.108	116.0

8.4 Sensitivity to detector parameters

On the common VIX sample, the diagnostic window is varied across 50, 63, and 76 trading days, and the warning threshold across 1.6, 2.0, and 2.4. The historical standardization window and all event-matching rules remain fixed. Across the nine specifications, recall ranges from 45.5% to 65.5% and precision from 9.1% to 11.5%. Raising the threshold generally lowers recall and raises precision, as expected. Changing the window does not eliminate the signal, although the exact tradeoff varies.

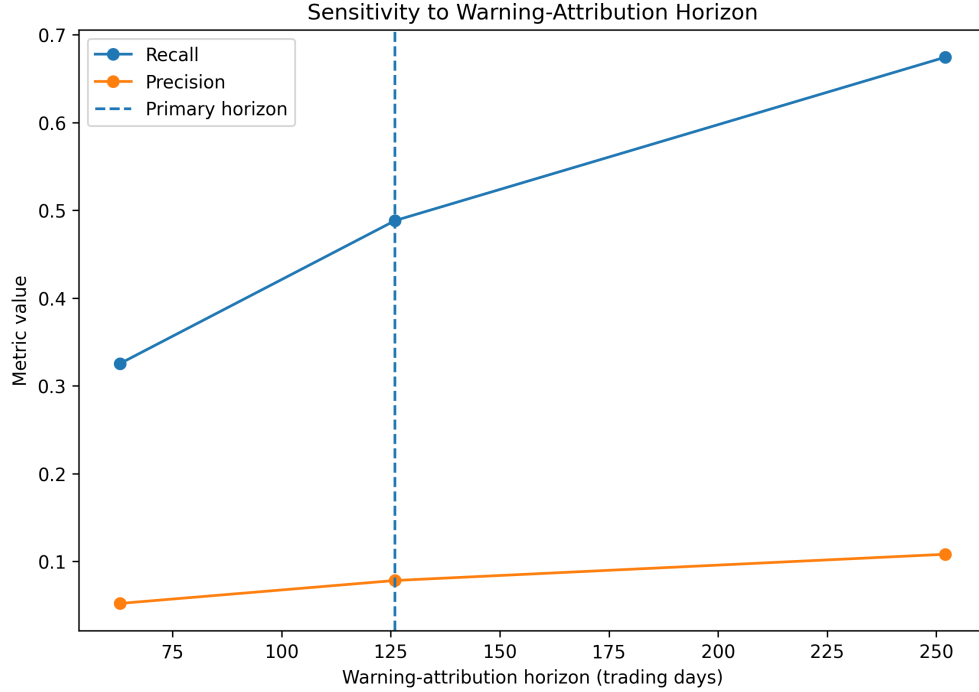


Figure 9: Sensitivity of recall and precision to the warning-attribution horizon.

Table 10: Detector-parameter sensitivity on the common sample

Window	Threshold	Recall	Precision	Median lead
50	1.6	0.655	0.094	82.0
50	2.0	0.564	0.091	76.0
50	2.4	0.491	0.098	73.0
63	1.6	0.655	0.107	89.0
63	2.0	0.564	0.104	83.0
63	2.4	0.491	0.114	63.0
76	1.6	0.582	0.101	83.5
76	2.0	0.545	0.107	60.5
76	2.4	0.455	0.115	62.0

9 Discussion

The evidence offers three main lessons.

First, changes in factor-return tail shape contain some advance information about mechanically defined factor collapses. The frozen excess-kurtosis detector identifies nearly half of holdout events, and successful warnings lead collapses by a median of roughly one quarter. Performance is not confined to Momentum: SMB and RMW exhibit the highest descriptive recall.

Second, the signal is weak relative to the volume of alarms. More than nine in ten warning

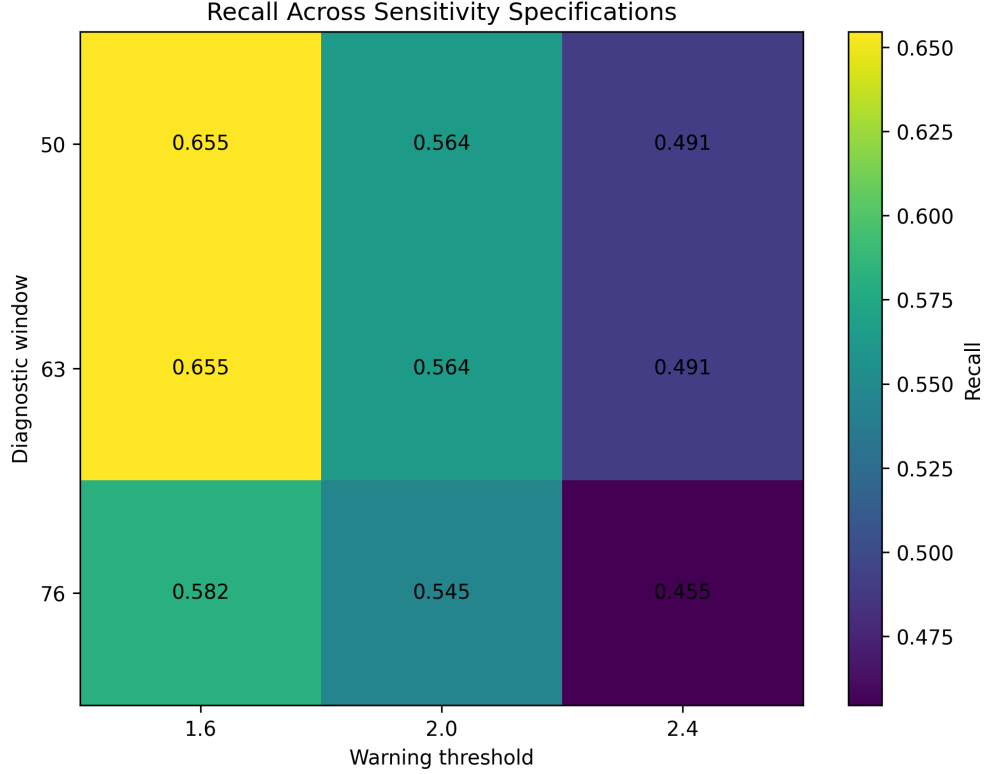


Figure 10: Recall across diagnostic-window and warning-threshold sensitivity specifications.

events are unmatched under the primary horizon. The exploratory warning-classification appendix confirms that most unmatched warnings are not simply slightly too early; 377 of 536 total warnings occur more than one trading year before the next collapse. Excess kurtosis often identifies unusual return distributions that do not develop into a collapse under the chosen Sharpe-based definition.

Third, broad market stress and factor-specific tail instability are related but not interchangeable. VIX achieves higher common-sample recall but generates far more factor-specific alarms and lower precision. The nontrivial counts of kurtosis-only and VIX-only detections suggest a future multivariate system could be more informative than either signal alone. Such a system would require a new development and holdout design rather than post-hoc combination on the current sample.

The statistical results impose an important limit on interpretation. Observed performance is better than the average circularly shifted warning pattern but not by a statistically significant margin. The detector should not be described as validated, predictive in a strong statistical sense, or ready for investment deployment. A more appropriate conclusion is that the study finds a modest, economically interpretable association that survives honest holdout evaluation but remains too noisy and underpowered for strong inference.

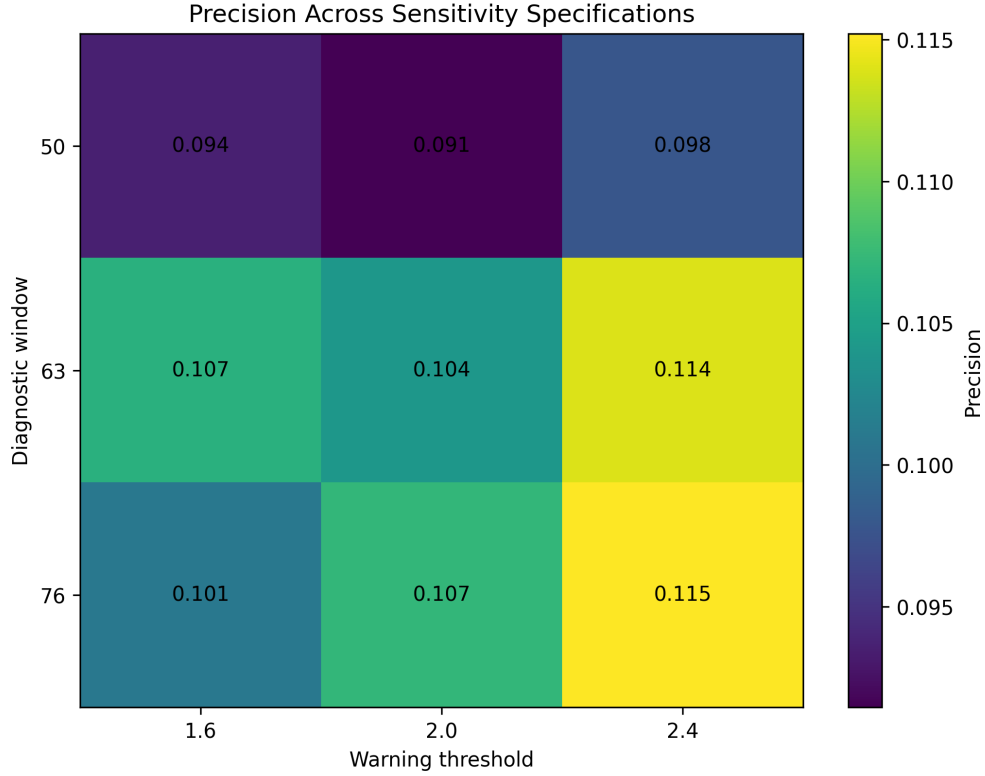


Figure 11: Precision across diagnostic-window and warning-threshold sensitivity specifications.

10 Limitations

Collapse-definition dependence. The ground truth is generated from a trailing 252-day Sharpe ratio and a 1,260-day reference distribution. This definition emphasizes sustained deterioration and can date a collapse long after the first economically salient crash. The October 2009 Momentum development date illustrates this issue. Alternative objective definitions based on drawdowns, cumulative losses, or regime transitions could produce materially different labels.

Small effective sample. Although daily data are abundant, the relevant sample size is the number of collapse episodes, not the number of return observations. There are only 86 holdout events and 11–17 per factor. Factor-level confidence intervals are therefore wide, and null results can reflect either weak effects or low power.

Event dependence. Collapse events and warnings may cluster around common macroeconomic regimes. The bootstrap resamples within factor and does not fully model cross-factor or serial dependence. The circular-shift null preserves within-factor warning spacing but may relocate warnings across structurally different eras.

Factor selection and survivorship. The six factors are established academic factors with long histories. This choice improves reproducibility but may not represent the broader universe of proposed premia. Studying factors that survived into standard data libraries may induce a form of selection bias.

Development-case mismatch. The pre-registered historical development context was the 2008–2009 Momentum crash, but the objective rule dates the nearby sustained collapse in October 2009. The detector is therefore developed against a post-crash deterioration label rather than the main crash onset. This mismatch limits how directly the result speaks to abrupt momentum crashes documented in the literature.

Warning-event definition. Precision depends on how consecutive or nearby warning days are grouped. The five-day reset rule is frozen and transparent but not uniquely correct. Similarly, the one-to-one matching convention counts additional warnings before an already detected collapse as false positives in the primary analysis.

No economic utility test. The study is intentionally not a trading-strategy backtest. It does not account for turnover, position sizing, transaction costs, utility, or the cost of reducing factor exposure during false alarms. Moderate recall and low precision could still be useful in some risk-management settings, but that cannot be inferred from the present classification metrics alone.

11 Conclusion

This paper conducts a pre-specified, repository-committed, causal holdout evaluation of whether factor-return distribution diagnostics warn before equity factor premia collapse. Of three candidate diagnostics developed on a single Momentum episode, only rolling excess kurtosis provides a qualifying advance warning at more than one adjacent threshold. The frozen detector identifies 42 of 86 holdout collapses with a median lead time of 66 trading days, but its 7.8% warning precision indicates a severe false-alarm problem.

The statistical evidence is modest. Observed recall and precision exceed the means of a circular-shift null but yield one-sided p -values of 0.164, and no factor-level result survives multiple-testing correction. On a common post-1990 sample, the detector is more selective than a causal VIX baseline but has lower recall. Kurtosis and VIX identify different subsets of collapses, suggesting—but not statistically establishing—that factor-specific tail instability may contain information distinct from broad market volatility. The result is descriptive rather than a validated combined model.

The most defensible interpretation is therefore neither success nor failure in a binary sense. Excess kurtosis is a plausible candidate fragility indicator: it generalizes across factors, provides meaningful lead time when correct, and is reasonably stable to moderate parameter changes. At the same time, it is noisy, statistically inconclusive, and unsuitable as a standalone collapse classifier. Future work

should test alternative objective event definitions, richer multivariate signals, dependence-aware inference, and genuine forward or rolling-origin evaluation on newly arriving data.

A Pre-Registration Summary

The original dated pre-registration is retained in the project repository and should be attached verbatim as supplementary material in any submitted version. The public repository should also identify the pre-registration commit and the frozen-parameter commit so that the timing of those decisions can be independently verified. Its core frozen commitments were:

- six daily U.S. factors: MKT–RF, SMB, HML, RMW, CMA, and MOM;
- one Momentum development episode and all other events as holdout;
- a 252-day rolling Sharpe collapse measure with a prior 1,260-day reference distribution;
- a two-standard-deviation collapse boundary with five-day entry and exit confirmation;
- three candidate diagnostics: skewness, excess kurtosis, and squared-return autocorrelation;
- trailing, causal-only computation with one-day-shifted historical normalization;
- a 126-trading-day warning horizon and five-day warning reset;
- reporting of lead time, precision, recall, false positives, failed diagnostics, and sensitivity results;
- Benjamini–Hochberg correction as the primary multiple-testing procedure, with Bonferroni as a stricter comparison.

B Exploratory Warning Classification

The primary analysis classifies all unmatched warning events as false positives. An exploratory appendix decomposes those warnings without changing the primary metrics. Among 536 warning events, 42 are primary true positives, 10 are additional warnings within the primary 126-day horizon of an already matched collapse, 23 occur 127–252 days before the next collapse, 377 occur more than 252 days before the next collapse, 18 occur during an active collapse, and 66 have no later holdout collapse before sample end.

Table 11: Exploratory warning-event decomposition

Classification	Warning count
Primary true positive	42
Duplicate within 126-day horizon	10
Extended early warning (127–252 days)	23
Too early to attribute (> 252 days)	377
Warning during active collapse	18
No later collapse before sample end	66

Only 14.0% of all warnings occur within one trading year before a collapse when primary matches, duplicates, and extended warnings are combined. This decomposition reinforces rather than overturns the primary low-precision result.

C Reproducibility and Software Checks

The empirical workflow is implemented in seven Google Colab notebooks:

1. data acquisition and cleaning;
2. objective collapse definition;
3. development-case diagnostic construction and parameter freeze;
4. holdout evaluation and warning-classification appendix;
5. bootstrap and permutation validation;
6. VIX baseline and sensitivity analysis; and
7. publication figures and tables.

Raw source files are retained with checksums, processed data are stored as Parquet files, parameters are saved in JSON, and result tables are saved as CSV. Causal-invariance tests alter future observations and verify that all earlier diagnostic outputs remain identical. The repository should retain the pre-registration commit and frozen-parameter commit in its public history.

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